

Time Value of Money & Simple Interest

Group #2

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Time Value of Money & Simple Interest

Time Value of Money

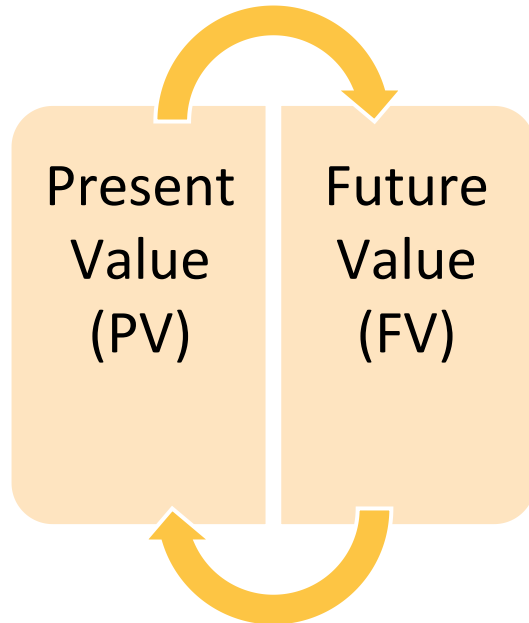
Core Concept of Time Value of Money (TVM):

Definition:

The value increase of money over time when invested or used in production.

“A dollar today is worth more than a dollar tomorrow because it can be invested to earn a return.”

Two Key Dimensions:



Present Value (PV) = current worth of future cash flow



Future Value (FV) = value of today's money after growing over time



Bridge: Discount rate / Interest rate

Engineering Economics Meaning:

- You cannot add cash flows from different years directly
- Must discount future amounts to **present value** using a discount rate

Why Money Has Time Value (4 reasons)

Inflation

"The purchasing power of money decreases over time due to inflation. \$100 today buys more than \$100 in the future."

Consumption Preference

"People generally prefer immediate consumption over delayed gratification. Present goods are valued more highly."



Opportunity Cost

"Future payments carry risk. Money received today is certain, while future payments may not materialize."

Risk & Uncertainty

"Money today can be invested to earn returns. Delaying receipt means losing potential earnings."

Together, these explain why future money must be "discounted."

Real-Life Examples with Data

Example 1 (Cash vs. Yu'E Bao)

With ¥10,000 at year start
Invest in Yu'E Bao (~1.05% annual return)

Strategy Choice	Principal Amount	Nominal Year-End Amount	CPI Inflation Adjustment (Dec 2025)	Actual Year-End Purchasing Power Equivalent
Strategy A: Cash Idle	10,000 CNY	10,000 CNY	-80 CNY (0.8% CPI)	9,920 CNY (actual depreciation)
Strategy B: Deposit in Yu'E Bao	10,000 CNY	approx. 10,105 CNY	-80 CNY (0.8% CPI)	10,025 CNY (real value increase)

Example 2 (Payment Choice)

Annual interest rate = 9%
You have \$100 today

Action	Result after 1 year
Keep \$100 at home	\$100 (no growth)
Deposit in bank	\$100 + \$9 interest = \$109

Conclusion:

- ✓ \$100 today is not equal to \$100 next year
- ✓ You would only wait if future amount > \$100 (e.g., \$109)
- ✓ The extra \$9 is the time value of money

02

Time Value of Money & Simple Interest

Simple Interest

Definition of Simple Interest

Simple interest is calculated only on the original principal amount. Interest does not compound.

Three essential components:

Principal (P)

- The principal is the base for interest calculation.
- I.E.:The initial amount you borrow or invest. For example, if you take out a loan of \$20,000, the principal is \$20,000.

Interest Rate (i)

- The interest rate is the percentage of interest charged or paid on the principal over a specific period of time. It determines the amount of interest and is typically expressed as an annual percentage rate (APR).

Time Period (n)

- Time Period is the duration for which money is borrowed or invested, typically measured in years. It determines how long interest will accumulate. A longer time period means more interest will be charged (for loans) or earned (for investments).

Simple & Compound Interest & Future Value Formula

Simple Interest

- Interest is calculated only on the initial principal.
- Formula: $I = P \times i \times n$.
- It is commonly used for short-term deposits or certain loans.

Compound Interest

- Interest is added to the principal at the end of each compounding period, and in the next period, interest is calculated on the new total (principal plus interest)
- Formula: $F = P \times (1 + i)^n$

Future Value Formula

- Future value represents the total amount of money due at the end of a loan period or the total amount accumulated in an investment
- Formula: $F = P + I = P(1 + i \times n)$
- It includes both the original principal and all interest earned.



Example Calculation of Simple Interest

Textbook Example (3-3):



☐ Lend \$5,000 to a friend for 5 years at 8% simple interest per year.

☐ Interest = $\$5,000 \times 0.08 \times 5 = \$2,000$

☐ Total owed after 5 years = $\$5,000 + \$2,000 = \$7,000$

☐ Annual interest is constant at \$400.

Real-life loan payment breakdown:



☐ A \$50,000 loan at 8% for 3 years (simple interest):

☐ Month 1: ~\$333 interest + ~\$1,389 principal

☐ Month 18: ~\$183 interest + ~\$1,539 principal

☐ Month 36: ~\$11 interest + ~\$1,711 principal

Making early payments saves significant interest.

Key Characteristics of Simple Interest

Advantages

Simple and transparent calculation

Borrower does not pay interest on interest (no compounding)

Principal portion of each payment increases over time

Disadvantages

Lenders rarely use simple interest (they lose compound earnings)

Much lower returns than compound interest for long-term investments

Most real-world transactions (credit cards, mortgages, student loans) use compound interest

Real-World Applications:

- Traditional term loans
- SBA loans
- Equipment financing
- Working capital loans
- Commercial real estate loans
- Business lines of credit

Albert Einstein:

Understanding simple interest is the foundation for mastering compound interest. But for real wealth building, compound interest is “the eighth wonder of the world”

03

Time Value of Money & Simple Interest

Cash Flow Diagram

Definition of Cash Flow Diagram

A cash flow diagram is a summary of the costs and benefits of an engineering project over its life cycle, plotted against time. It visually shows the magnitude, sign (positive or negative), and timing of cash flows.

4 rules

Time axis

- First draw a horizontal time axis divided into equal time units, such as years, months, or quarters.

Arrows

- At each time point where a cash flow occurs, draw a vertical arrow.
 - A downward arrow represents a cost or expense (negative cash flow).
 - An upward arrow represents a receipt or income (positive cash flow).

Proportional drawing

- Arrows are generally drawn in proportion to the magnitude of the cash flow.

Default timing

- Unless otherwise stated, cash flows are typically assumed to occur at the end of each period.
 - Time 0 represents the present; time 1 represents the end of the first period.



Borrower vs Lender



Seller/Lender :

When you lend money or sell equipment, cash flows in (income), so it is positive.

Buyer/Borrower :

When you borrow money or purchase equipment, cash flows out (expense), so it is negative.

Perspective determines sign:

The positive or negative sign of a cash flow depends on your standpoint.

Mirror image:

One person's cash outflow is another person's cash inflow.



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DATA ANALYSIS REPORT

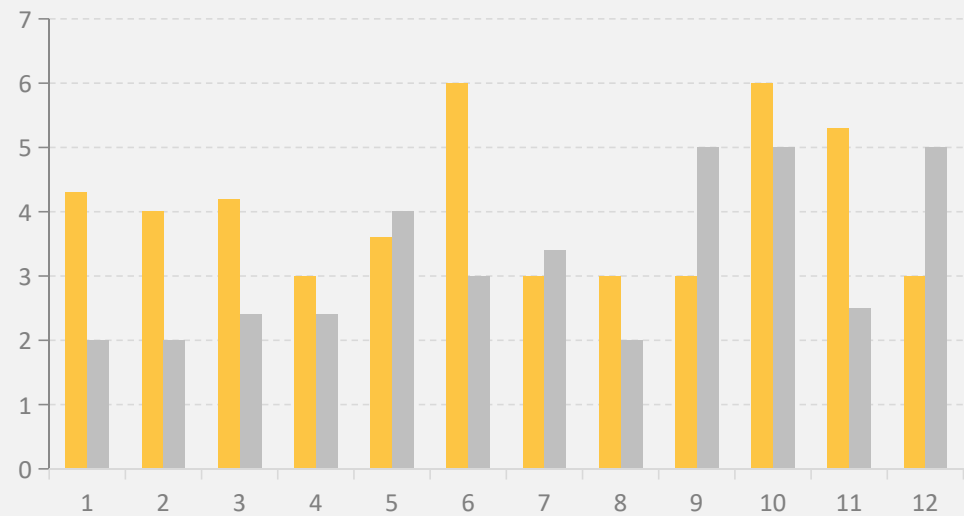
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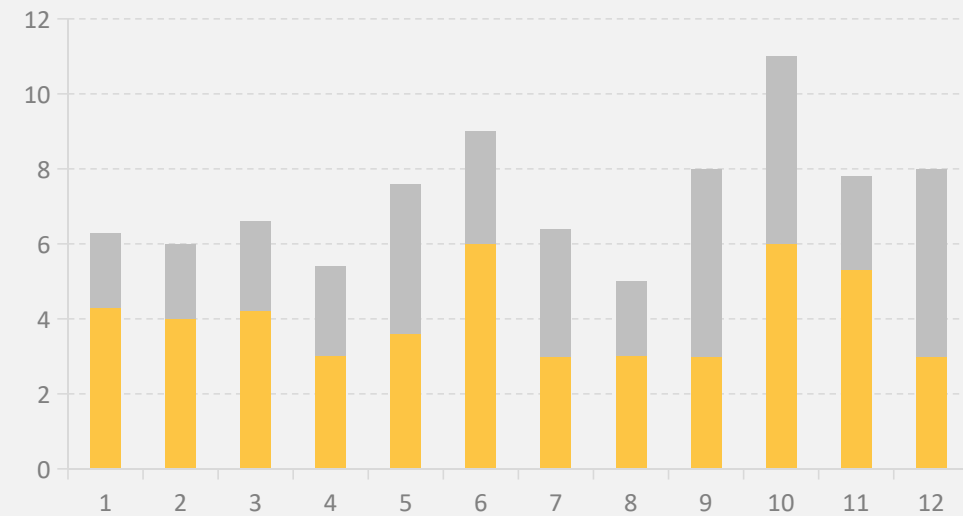
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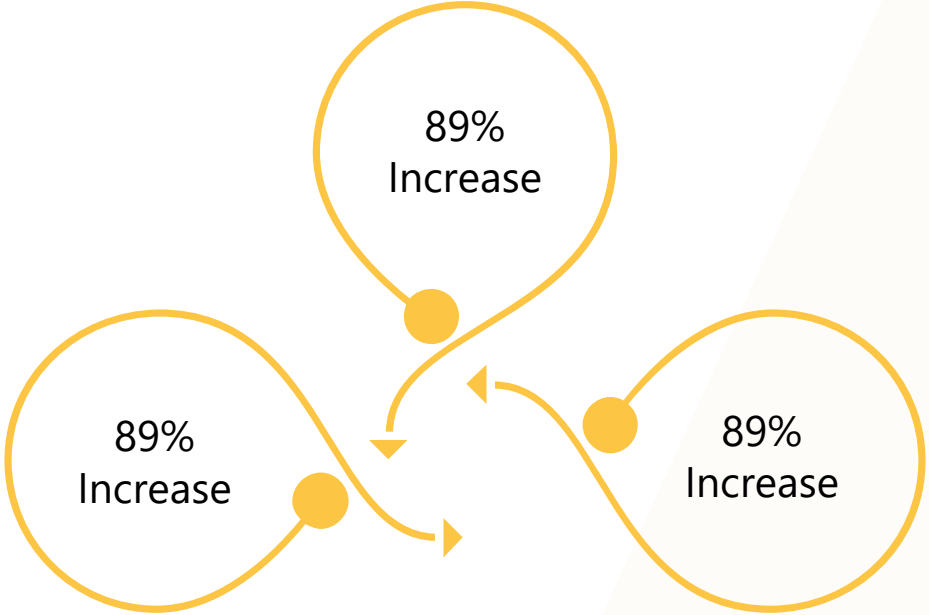


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